

Dynamics and Forces Test Results

6/4/2026

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1 Dynamics Test Results

1.1 5/17/2026 Testing (Pre TRR)

These tests were conducted with `dynamicsV4.m`. These test results with some damping were shown at TRR. This damping was removed for this analysis. Mission data compared against is from **4/17/2026 Edmonds Dive 5**. These tests only contains a **60 sec window** of data.

Dive Phase of Edmonds Dive 5 with `dynamicsV4.m`

Roll and pitch rates were unstable in this test. Roll rate diverged heavily, trending towards 2000 deg/s at the end of the 60s window. The pitch rate had both high and low frequency modes in its oscillation, with the instability worsening after the 20s mark. The heading rate of change spiked heavily at this point, peaking at over 10 times the nominal maximum deviations. The effects of these oscillations are shown clearly in the roll and pitch angles. The heading takes a little interpretation, with its discontinuities at 360 degrees and zero degrees, with the heading resetting as it passes that threshold. Simulated depth starts diverging around 5 seconds from the mission data indicating instability. The vertical velocity is oscillating at very high amplitudes compared to mission data, and it has high noise.

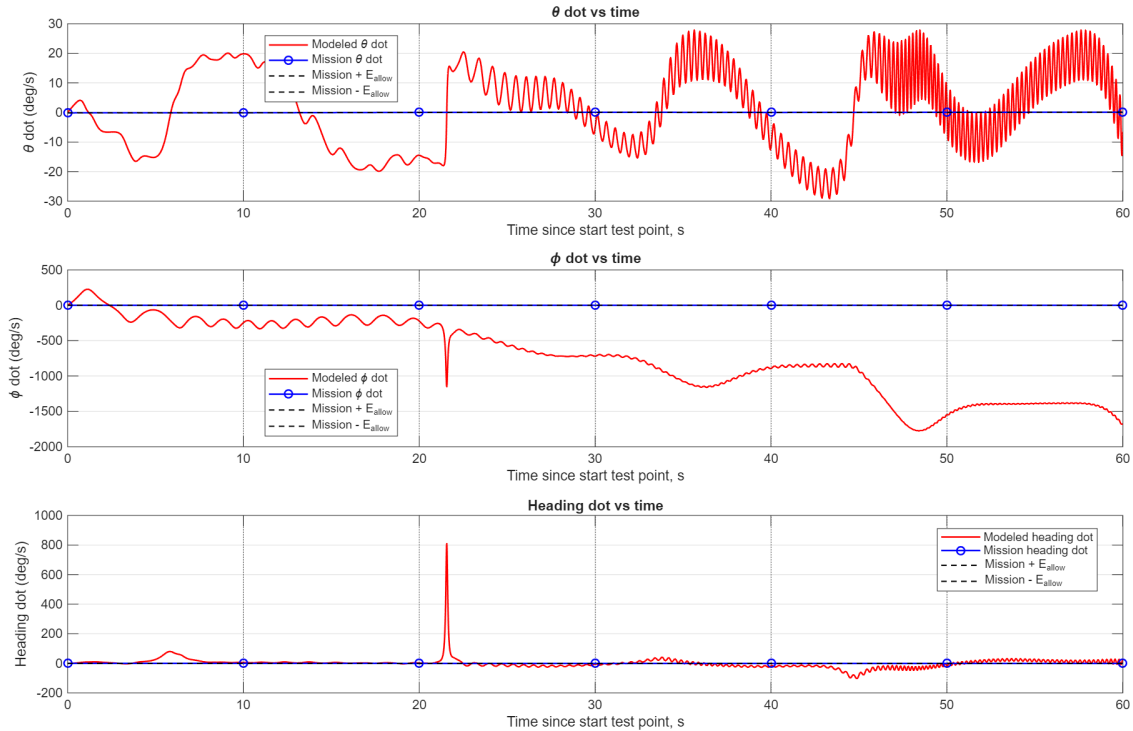


Figure 1: Angular rates vs time from dive phase of Dive 5 with `dynamicsV4.m`

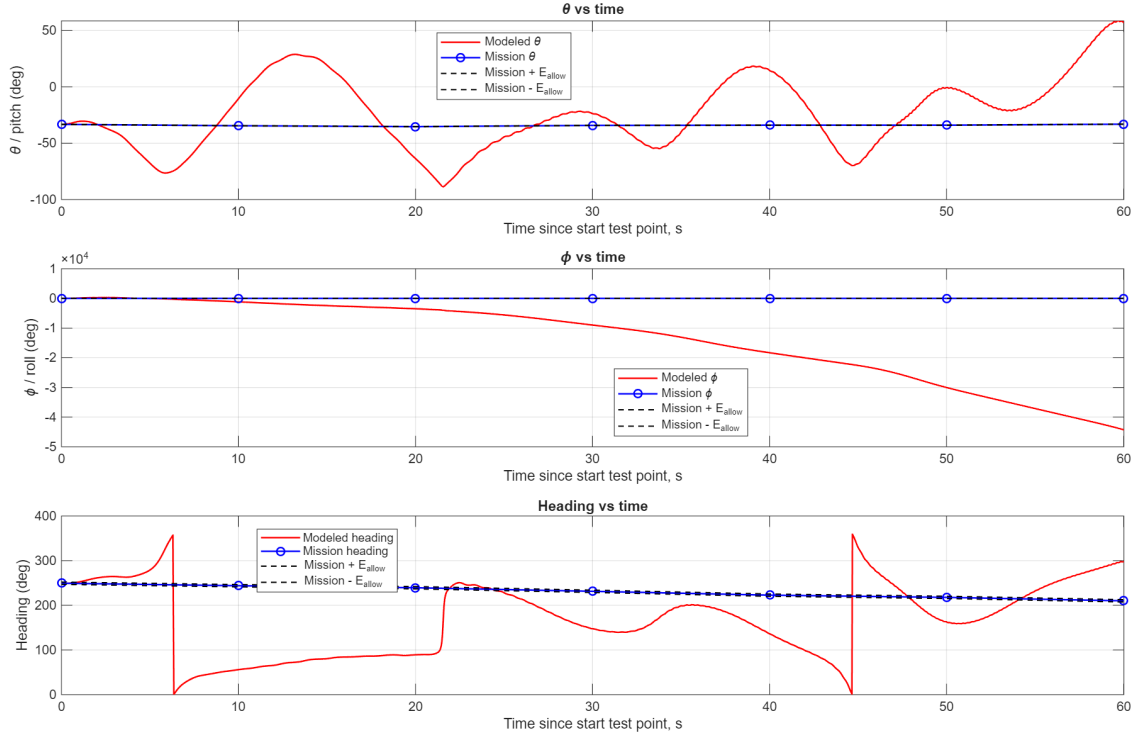


Figure 2: Euler angles vs time from dive phase of Dive 5 with dynamicsV4.m

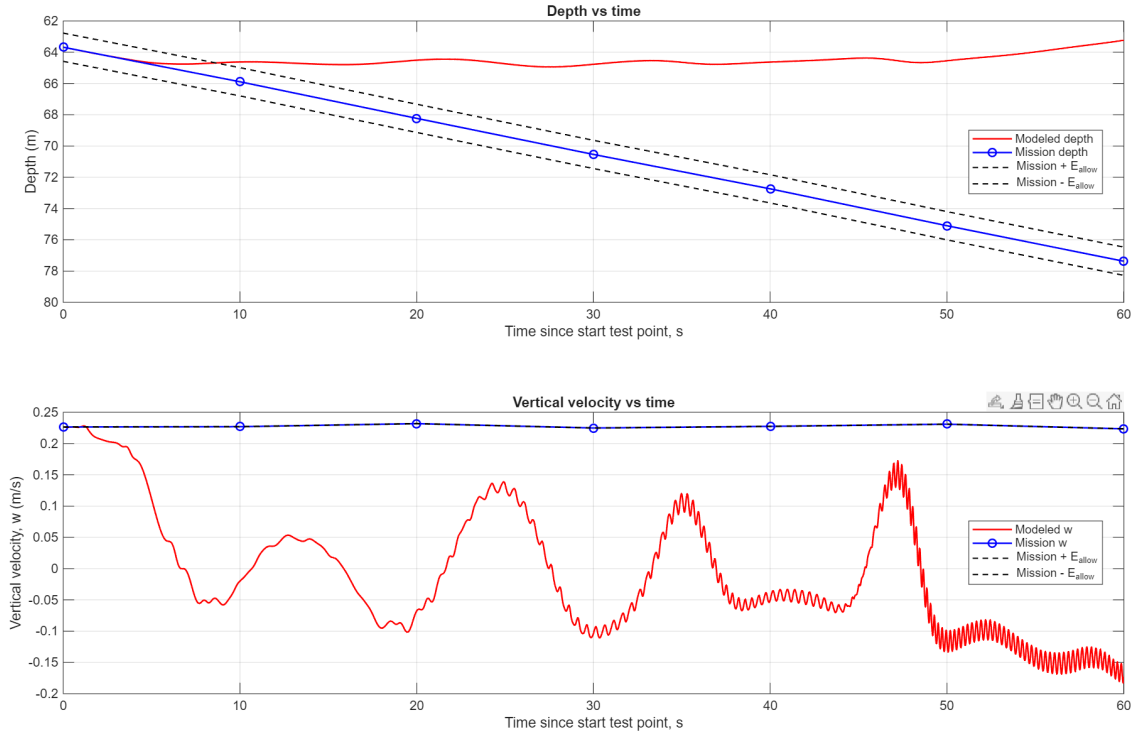


Figure 3: Depth and vertical velocity vs time from dive phase of Dive 5 with dynamicsV4.m

Climb Phase of Edmonds Dive 5 with dynamicsV4.m

Similar angular rates and attitude were found in the ascent phase. The pitch and roll have a similar amount of noise. Roll rate does not continue to diverge, staying at a constant roughly 300 deg/s. The heading rate has a similar, but less extreme, spike at the 20 second mark.

The vertical velocity continues to steadily diverge, with a 300% difference at the end of the window. The depth reflects this.

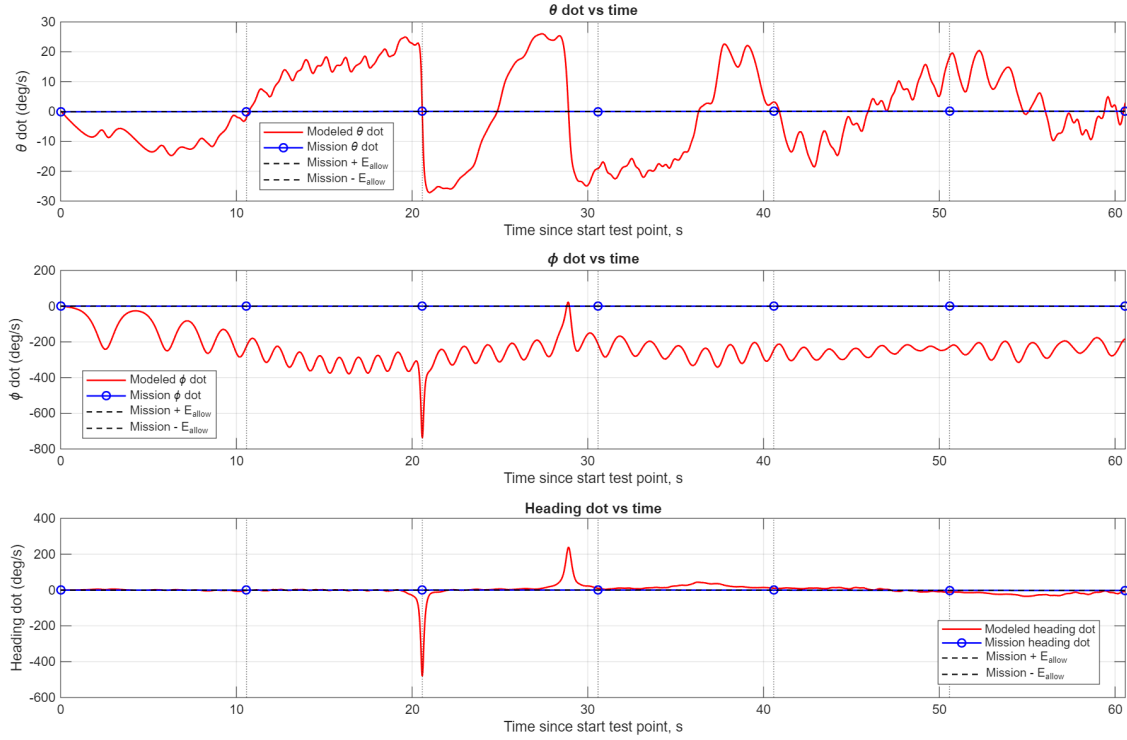


Figure 4: Angular rates vs. time from climb phase of Dive 5 with dynamicsV4.m

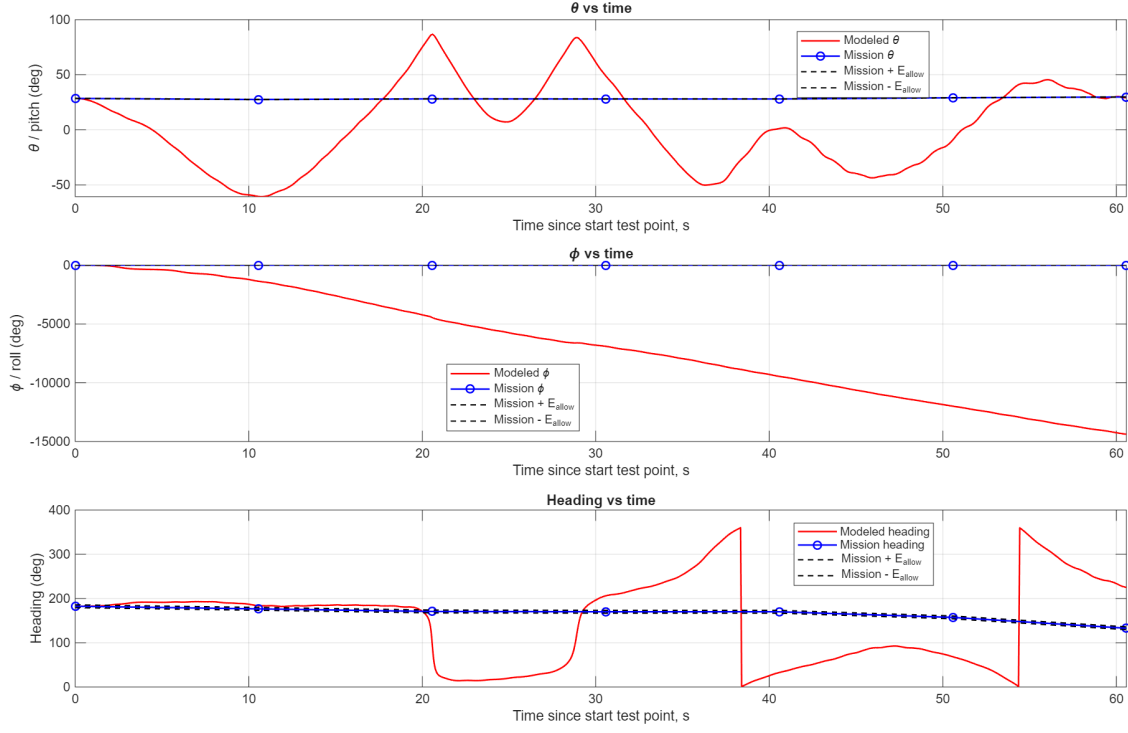


Figure 5: Euler angles vs time from climb phase of Dive 5 with dynamicsV4.m

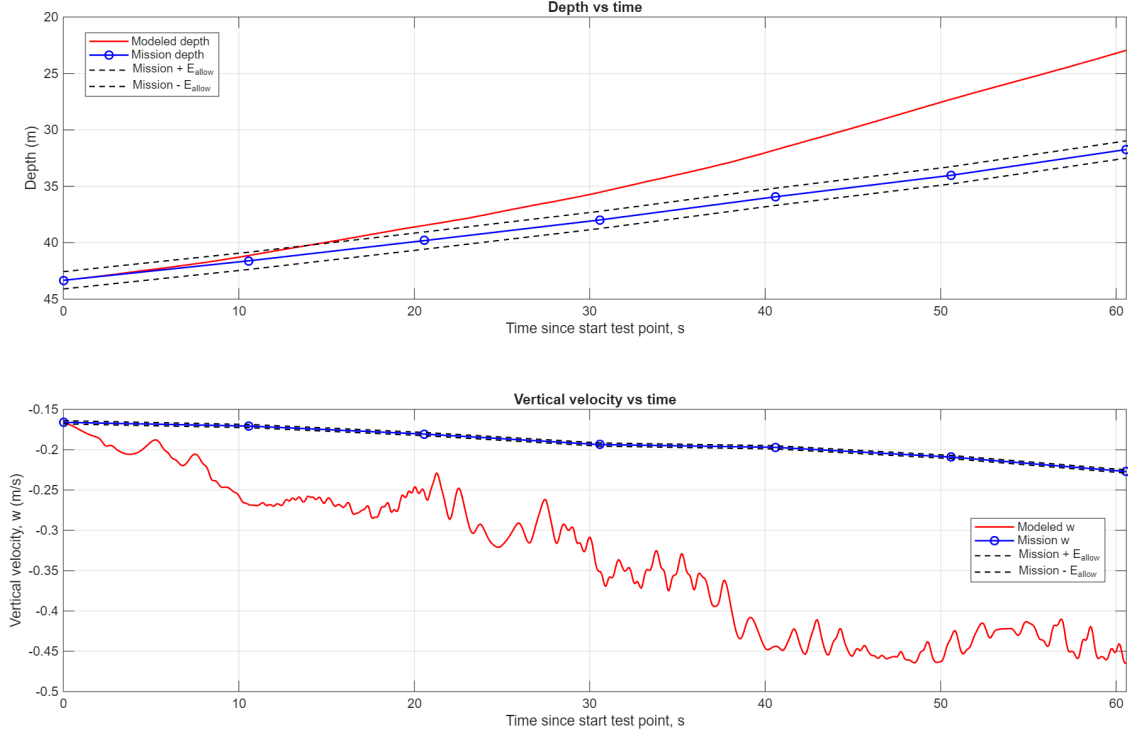


Figure 6: Depth and vertical velocity vs time from climb phase of Dive 5 with dynamicsV4.m

1.2 6/3/2026 Testing (Post TRR)

These tests were conducted with `dynamicsV6.06.m` after TRR. Mission data compared against is from **4/17/2026 Edmonds Dive 5**. These tests only contains a **60 sec window** of data. Wind tunnel data was not used.

The primary changes to the dynamics function were as follows: input forward and lateral velocities were properly converted to the glide path frame, the CG was no longer assumed to be at the CB, negative sideslip angles were properly mirrored in the coefficient table lookup, small typos were found and fixed, and sign errors in the battery CG location were fixed.

Dive Phase of Edmonds Dive 5 with `dynamicsV6.06.m`

Similar to the 5/17/2026 test, roll and pitch are both unstable. Pitch rate exhibits oscillatory behavior with a semi-constant magnitude of oscillation at around 20 deg/s. Roll rate displays growing unstable oscillatory behavior, diverging from the true point completely at around 47 seconds into the test. Heading rate exhibits some variability, with a spike at around 38 seconds, but continues to diverge away from the true line. Pitch displays oscillatory behavior, but with variability in the magnitude of said oscillations. Over the first 20 seconds of the test, the roll is relatively accurate, however at around the 20 second mark, it begins to oscillate around the true point, until a total divergence occurs at around 47 seconds. Heading exhibits a pattern of divergence, spike, divergence, localized roughly around the true value. The depth follows the trend of the true value for most of the elapsed test time, however diverges outside of the error bounds at around 54 seconds. The vertical velocity exhibits oscillatory behavior in a parabola pattern, with varying magnitudes of oscillations. Overall none of the data sets are within the error bounds.

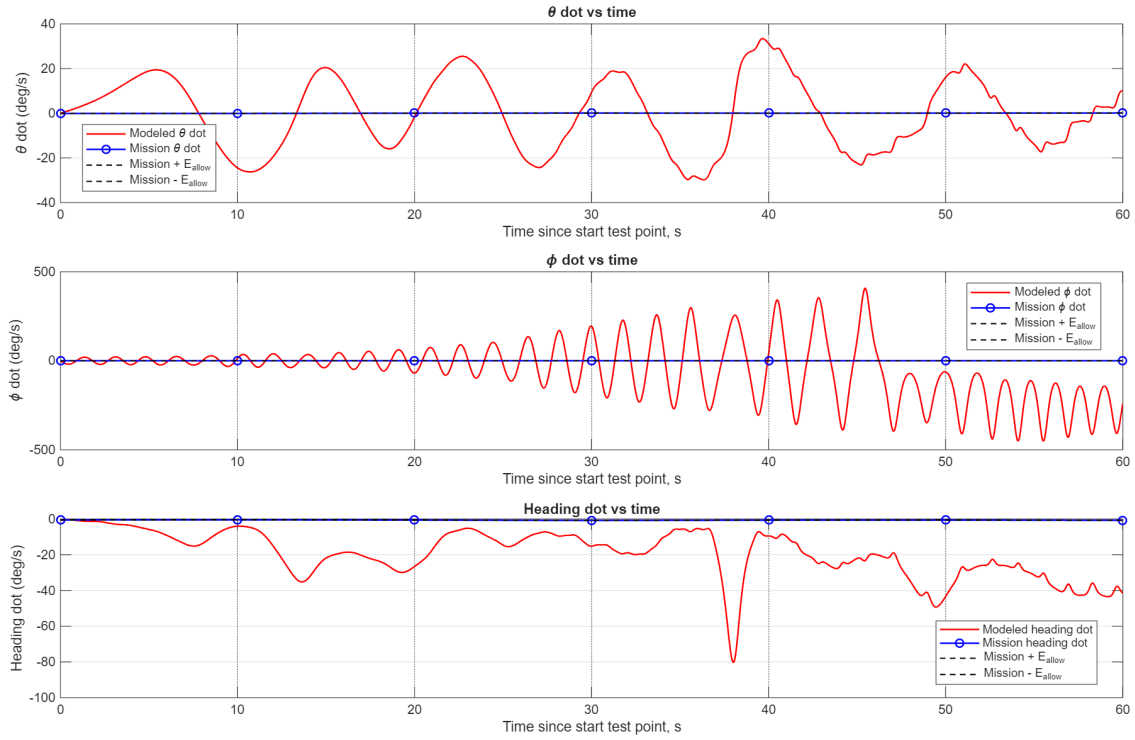


Figure 7: Angular rates vs time from dive phase of Dive 5 with dynamicsV6_06.m

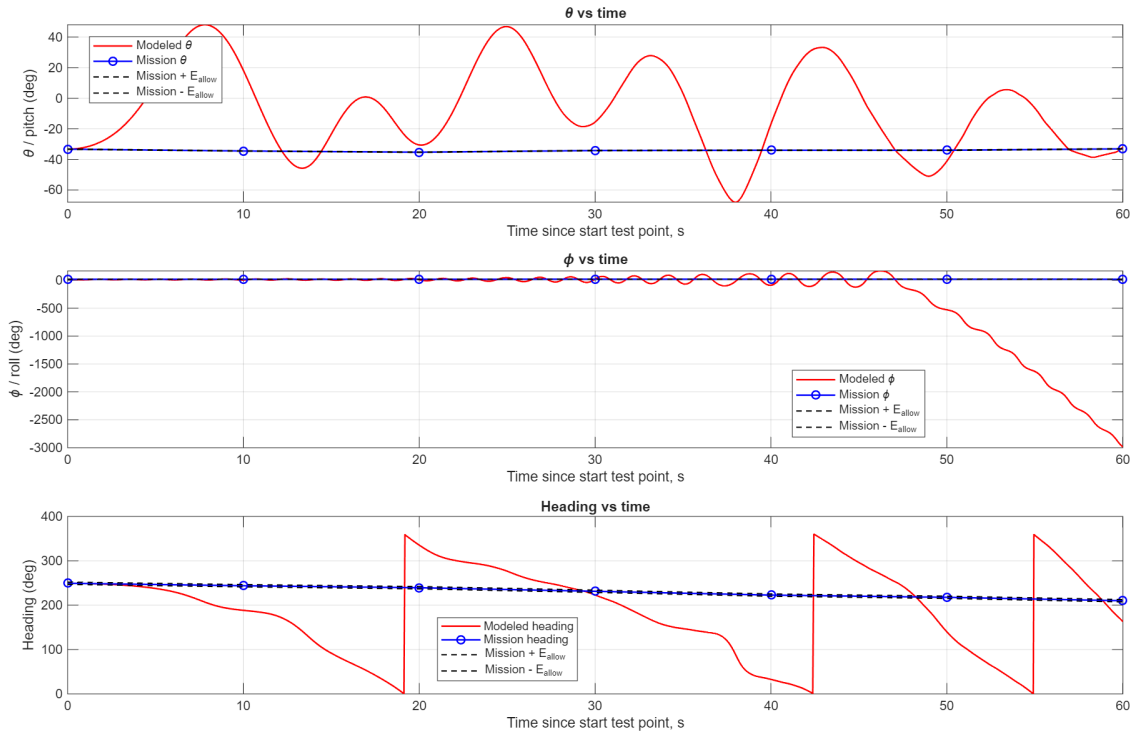


Figure 8: Euler angles vs time from dive phase of Dive 5 with dynamicsV6_06.m

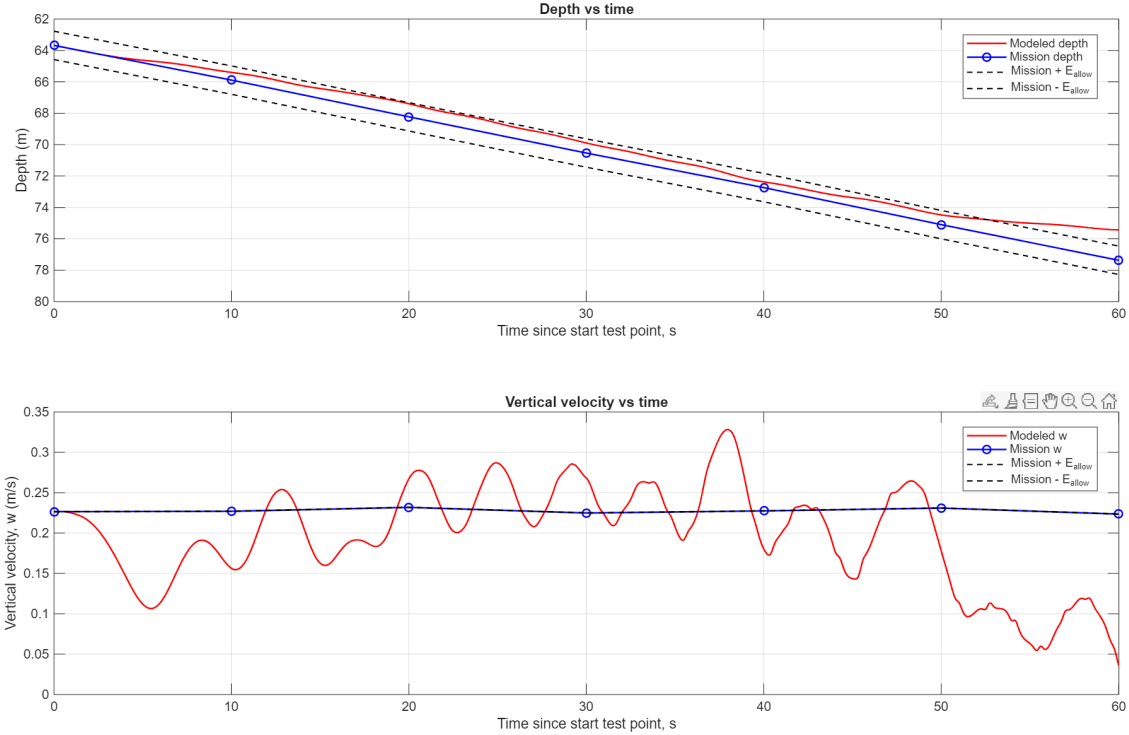


Figure 9: Depth and vertical velocity vs time from dive phase of Dive 5 with dynamicsV6_06.m

Climb Phase of Edmonds Dive 5 with dynamicsV6_06.m

Pitch rate exhibits oscillatory behavior, settling at a relatively constant magnitude of about 32 deg/s, centering around the true value. The roll rate also oscillates around the value, however its oscillations are widely variable. Heading rate also oscillates with variability, following a curved trajectory not centered on the true line. Pitch also oscillates, with an oscillation magnitude settling at around 30 deg, however not settling on the true line. Roll also oscillates widely, with a period of the test around the true line, but at around 54 seconds diverging again. Heading begins on the true line, before diverging steadily to a point, spiking at around 29 seconds, before displaying a wide parabolic trajectory. Similar to heading, depth begins within the error bars, before following a steady divergence pattern as the test goes on. Vertical velocity exhibits a oscillatory pattern following a diverging trajectory. Overall none of the data sets are within the error bounds.

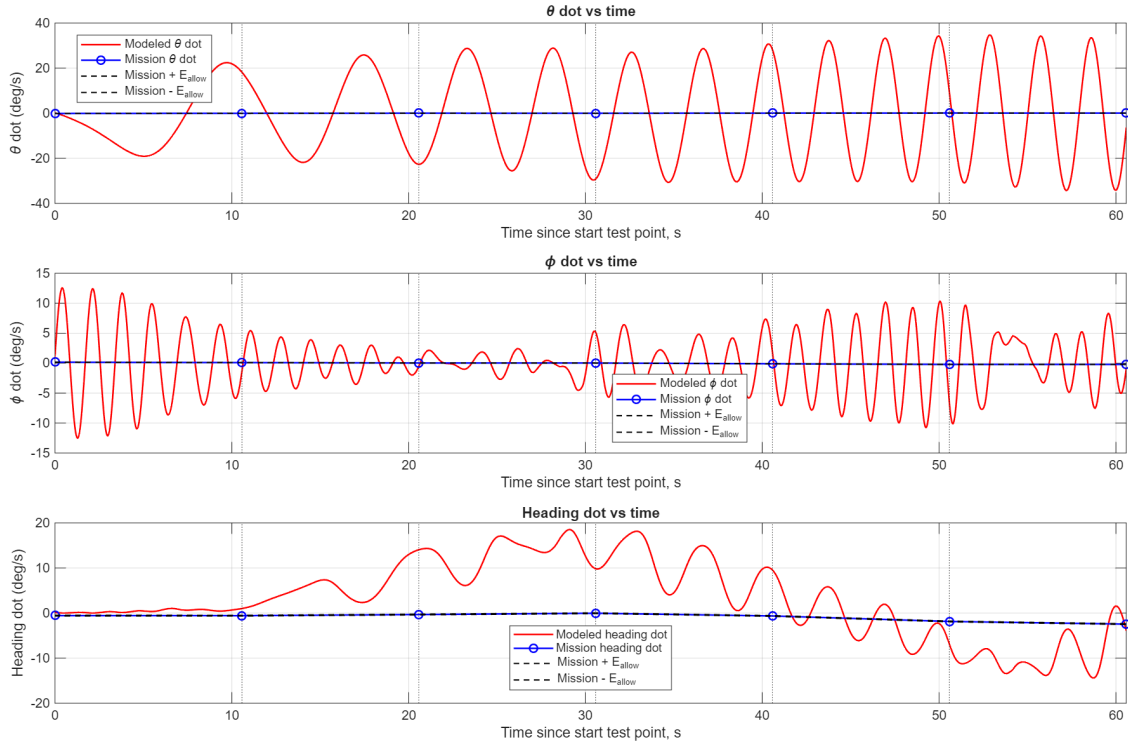


Figure 10: Angular rates vs time from climb phase of Dive 5 with dynamicsV6_06.m

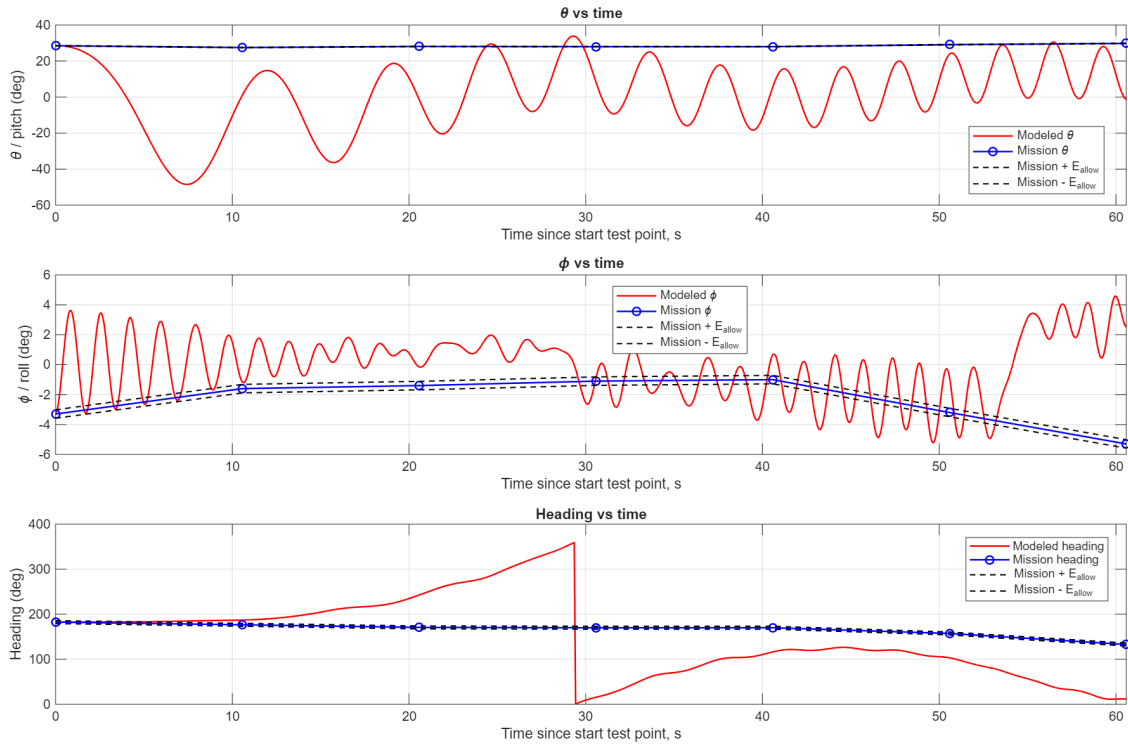


Figure 11: Euler angles vs time from climb phase of Dive 5 with dynamicsV6_06.m

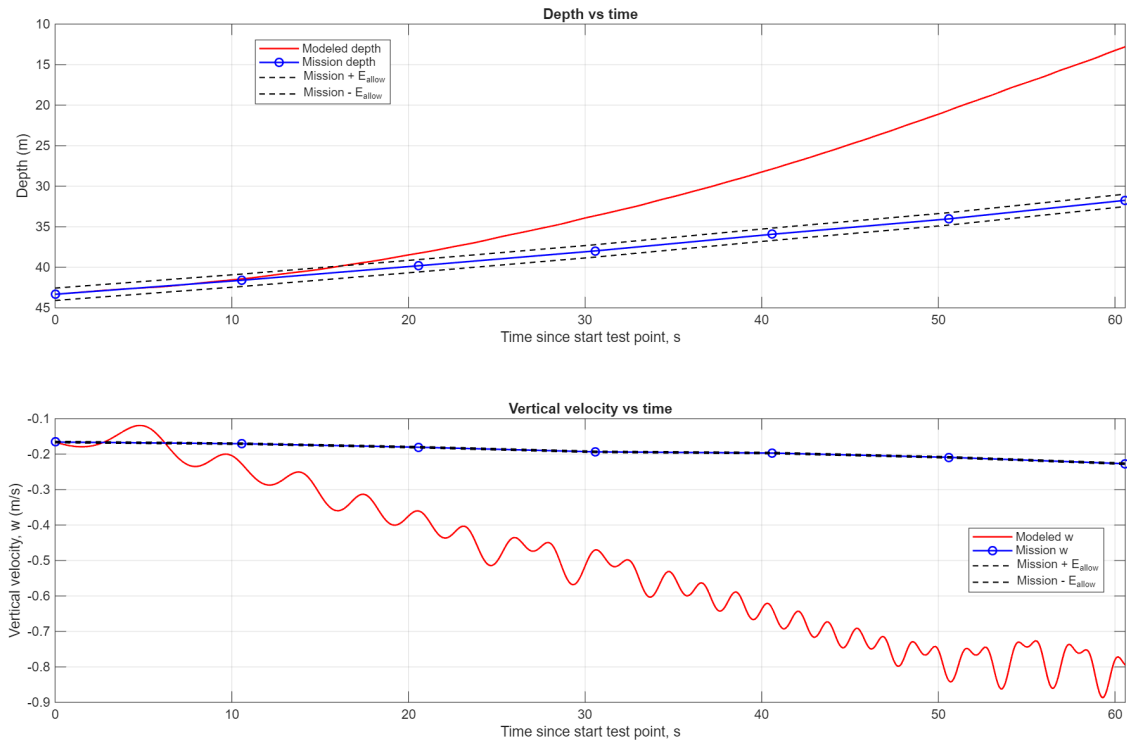


Figure 12: Depth and vertical velocity vs time from climb phase of Dive 5 with dynamicsV6.06.m

2 Forces Test Results

The forces on the glider were tested under the nominal (no damage) and planar (no side force) model. These forces (lift and drag, L and D respectively) were plotted against an angle of attack (AoA) sweep from $[-15, 15]$ degrees. The truth model used was the hydrodynamic model provided by Charlie Eriksen [1].

2.1 Pre-TRR Testing

These tests were conducted with computational fluid dynamics (CFD) simulations versus the hydrodynamic model (HDM). The results are as follows:

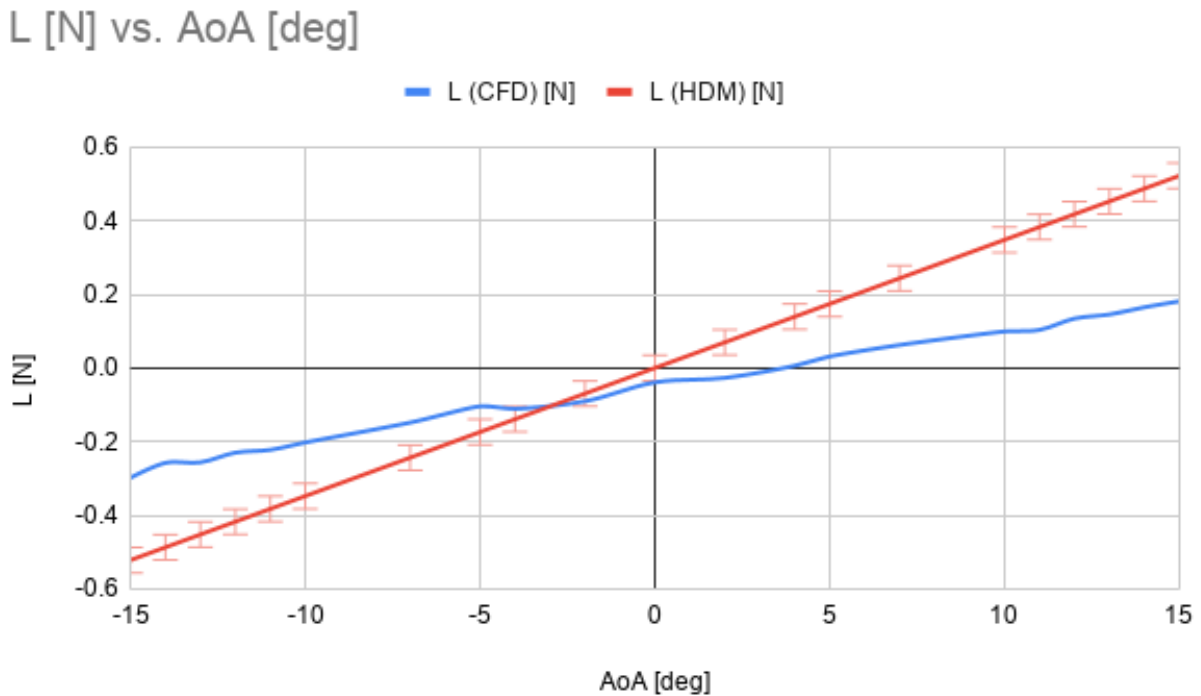


Figure 13: Lift force (N) from CFD and HDM versus AoA (deg)

Figure 13 shows the lift force calculated from the CFD simulation (blue) and the HDM (red) where the error bars are the allotted magnitude of ± 0.0344 N. The lift force from the CFD model only had two points that fall within the error bounds of the HDM and therefore fails the test. The two lines have the same trend, but the HDM is a perfectly linear and ideal model with zero lift at zero angle of attack.

D [N] vs. AoA [deg]

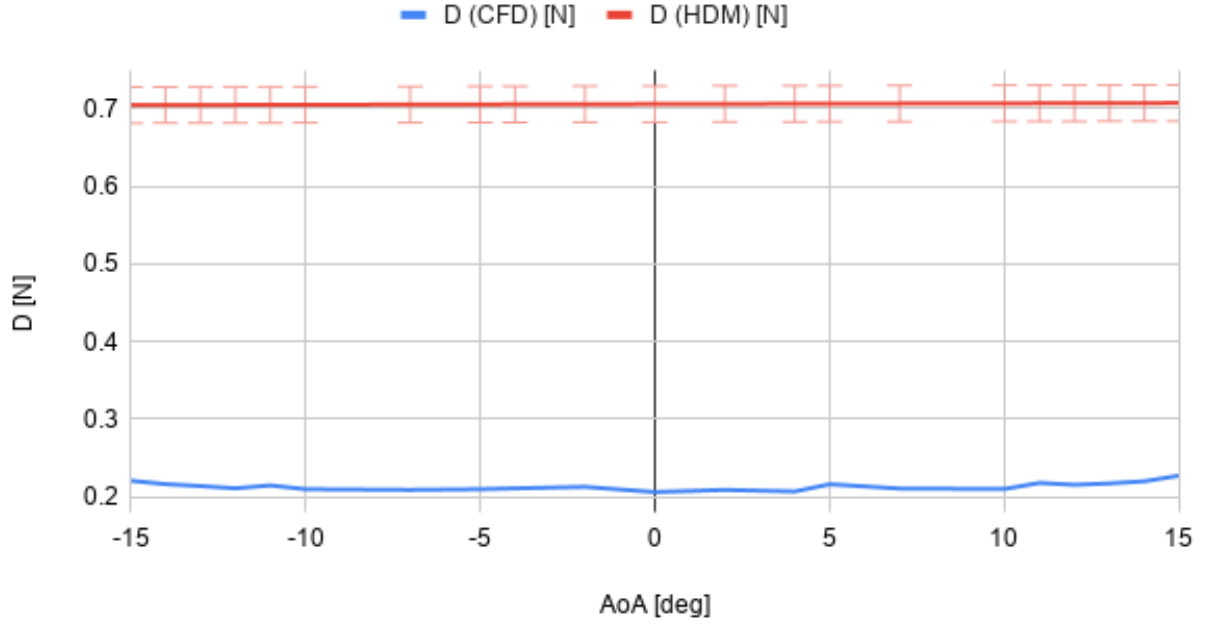


Figure 14: Drag force (N) versus AoA (deg)

Figure 14 shows the drag force calculated from both the CFD (blue) and HDM (red). The drag force from CFD does not fall within the error bounds of ± 0.0233 N and therefore fails the test. These results have the same trend, but are off by a magnitude of about 0.5 N.

2.2 Post TRR and Wind Tunnel Testing Results

The following show the test results comparing CFD, HDM, and wind tunnel (WT) force calculations versus an angle of attack sweep. The results are as follows:

L v AoA

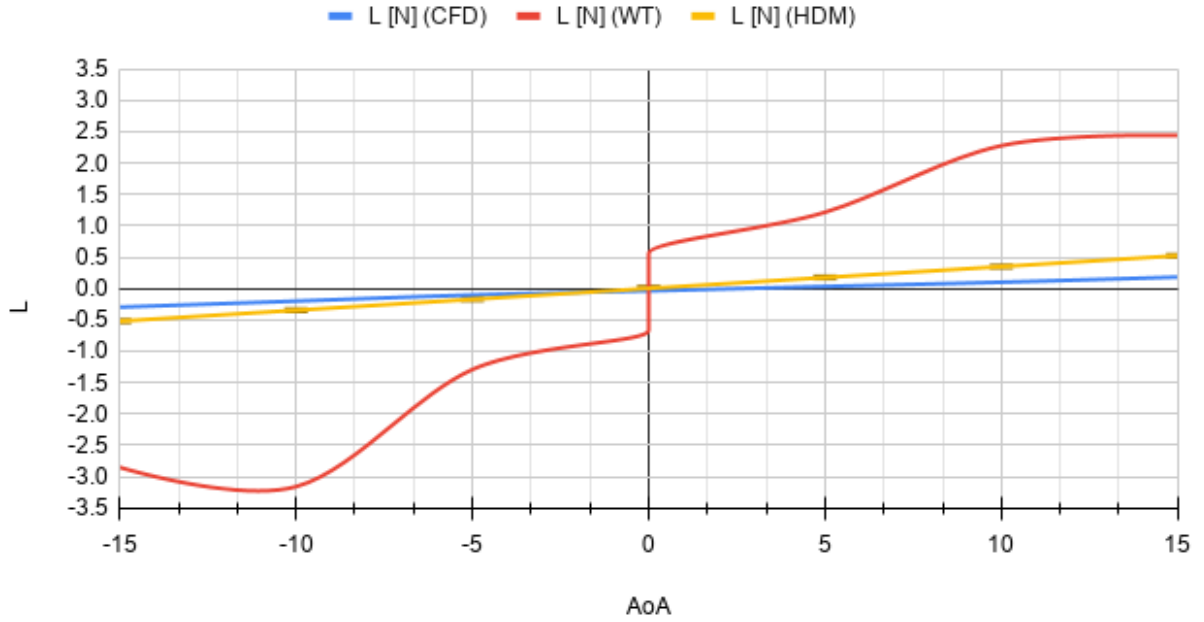


Figure 15: Lift force (N) from CFD, HDM, and WT against AoA (deg)

Figure 15 shows the lift from each model plotted against the angle of attack. The error bars are the same magnitude as the pre-TRR test. Both the CFD and wind tunnel results do not pass the test with the HDM being the truth model as there are only two points that fall within the error bounds provided. The wind tunnel results are more accurate to literature of lift along a nearly symmetrical surface, but even so the test still fails.

D vs. AoA [deg]

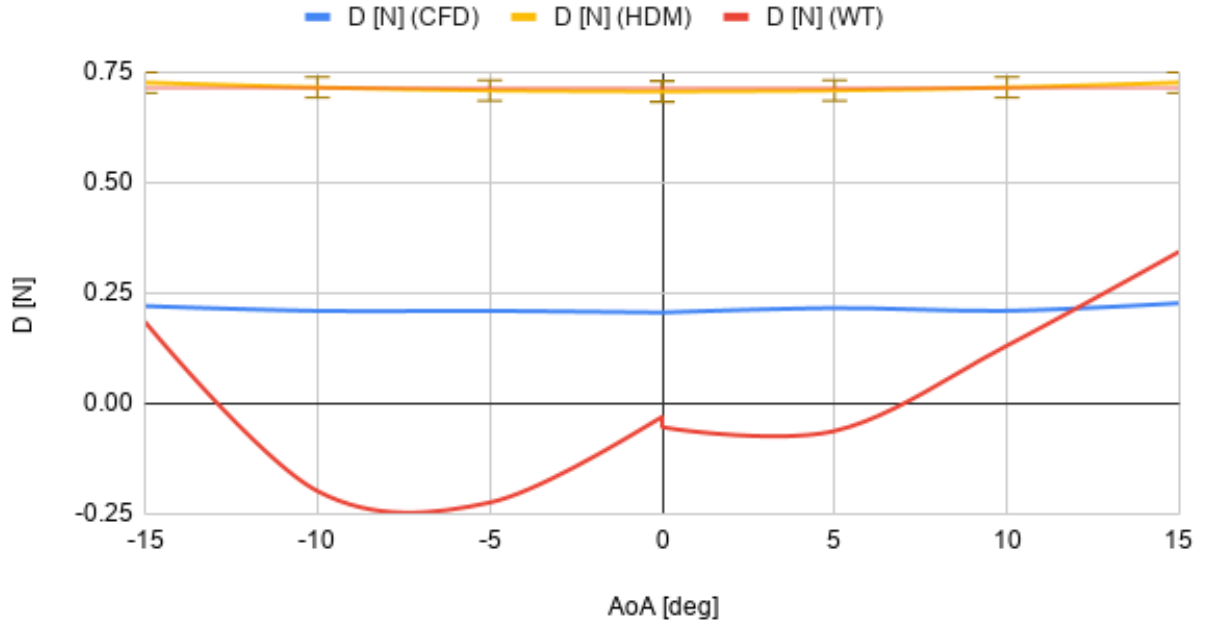


Figure 16: Drag force (N) from CFD, HDM, and WT against AoA (deg)

Figure 16 shows the drag from each model plotted against the angle of attack. The error bars are the same magnitude as the pre-TRR test. Both the CFD and wind tunnel results do not pass the test with the HDM being the truth model as no points fall within the error bounds. The CFD and HDM have the same trend again, but the wind tunnel model for drag is what is expected from literature.

The HDM is not an accurate truth model, but even if the truth model was switched to either CFD or wind tunnel the tests all fail. Further recommendations on how to improve future force models and test results are in the ROFASO suggestions document.

3 Other Dynamics Testing (not in TRR)

In order to test very basic dynamics, we isolated the forces, and ran the code for cases with known inputs and outputs.

3.1 Translation in z-direction using dynamicsV4.m

3.1.1 Test 1: Constant vertical velocity and no acceleration

Isolated forces and moments by setting ALL forces and torques ZERO. This includes moments cause by battery motion.

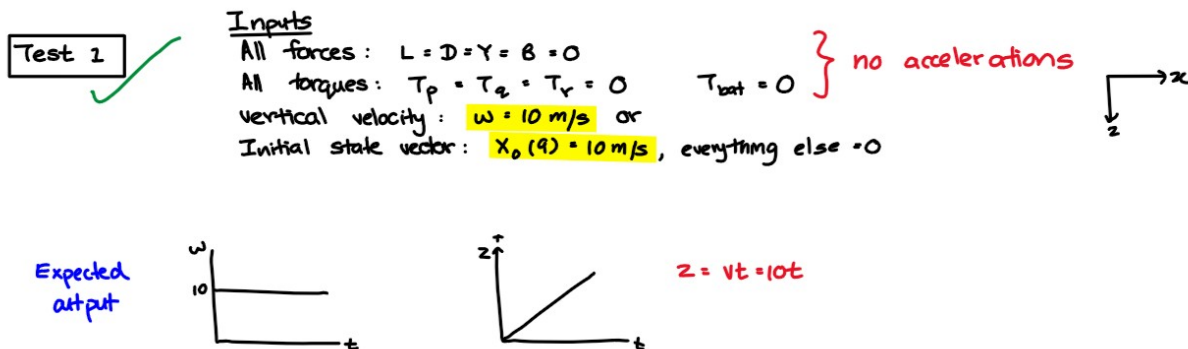
```

143 % Hydrodynamic forces in body frame
144 %L = q * CL * S; % lift
145 %D = q * CD * S; % drag
146 %Y = q * CY * S; % side
147 L = 0;
148 D = 0;
149 Y = 0;
150 F_hydro = [-D; Y; -L];

% buoyancy equation
%B = g_b * (m_total - rho * Volume);
B=0;

176 % Hydrodynamic moments (torques)
177 % Mp = q * Croll * b * S; % roll moment
178 % Mq = q * Cpitch * cbar * S; % pitch moment
179 % Mr = q * Cyaw * b * S; % yaw moment
180
181 Mp = 0;
182 Mq = 0;
183 Mr = 0;
184 Torques = [Mp; Mq; Mr];

```



Actual outputs:

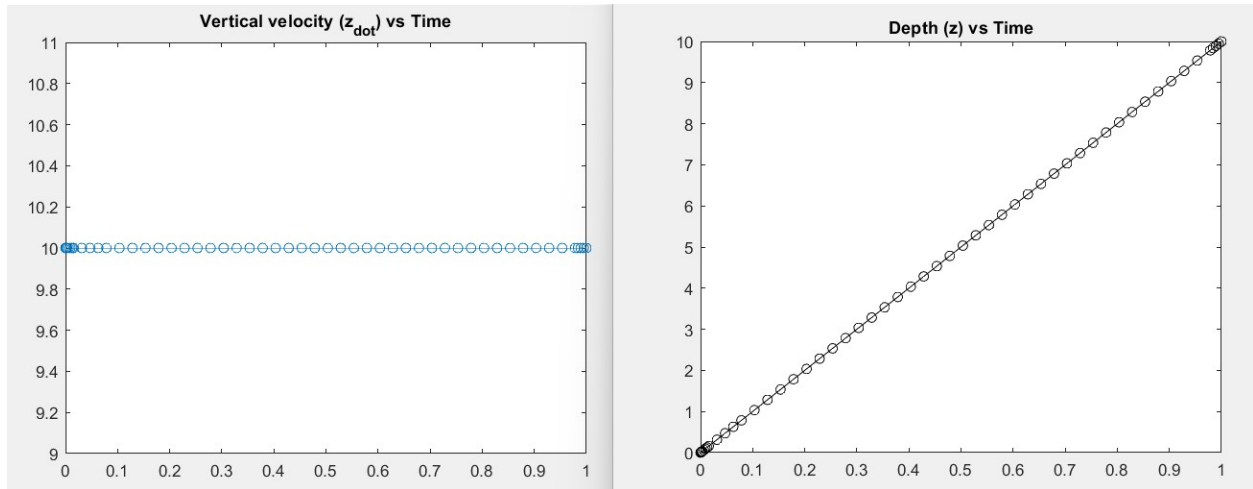


Figure 17: Vertical velocity or depth (vertical axis) vs. time (horizontal axis).

In this test we applied a constant vertical velocity with no acceleration. The test code used for this test can be found in the Appendix where all the inputs to the ODE function are listed.

When we apply a constant velocity with no acceleration, we expect the depth to be linear. Since the actual output matched these expected results exactly, this test passed.

3.1.2 Test 2: Constant acceleration in z-direction

Test 2 accel. in z direction 

Inputs: $L = -10 \text{ N}$
 $\omega = 10 \text{ m/s}$ $X_o(a) = 10 \text{ m/s}$
 other forces and torques = 0
 $\underline{u} = 0$
 $\underline{X}_o = 0$ except $X_o(a) = 10 \text{ m/s}$

→ What is acceleration due to this?
 from X_{dyn} , obtain accel. in z-dir. ($\dot{\omega}$)

X_{dyn} =

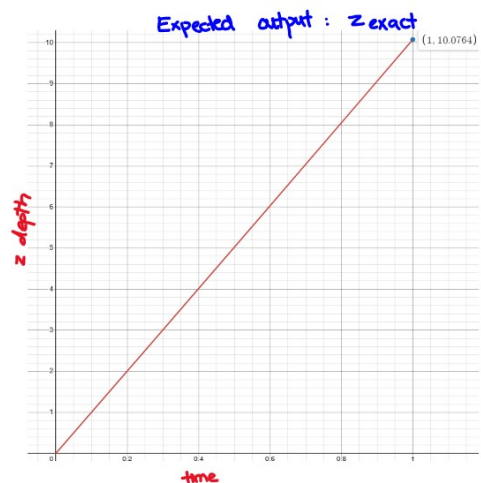
0
0
0
0
0
0
0
0
0
0.0764
0
0
10

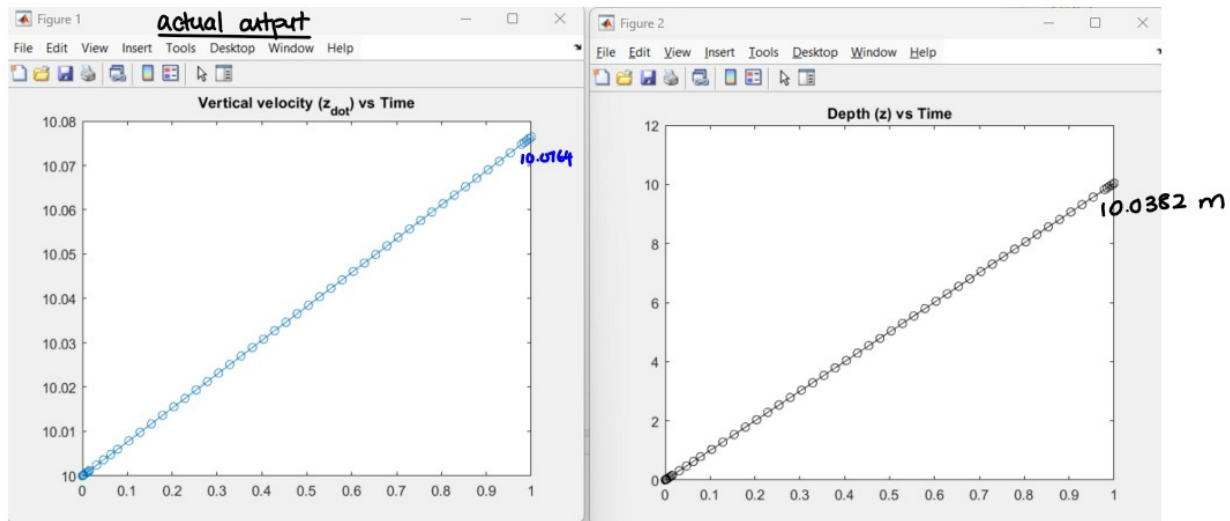
$\dot{\omega} = 0.0764 \text{ m/s}^2$

$Z_{\text{exact}} = 0.0764 t^2 + 10t$

$\Delta \omega = a t = 0.0764 t$
 $= 0.0764 \text{ m/s}$

$\star \omega$ is not updating w/ each time step in X_{dyn} .





$$\% \text{ error} = \frac{|10.0764 - 10.0382|}{10.0764} \times 100 = 0.38\%$$

Figure 18: Vertical velocity or depth (vertical axis) vs. time (horizontal axis).

In this test, a constant acceleration was applied in the z-direction by adding 10 N of lift. The expected velocity is linearly increasing which matches the actual output. The expected change in velocity at $t = 1 \text{ sec}$ is the acceleration times the time which is $0.0764 \text{ m/s}^2 \cdot 1 \text{ s} = 0.0764 \text{ m/s}$. This also matched the actual change in velocity. The expected final depth was 10.0764 m. However, the actual final depth in this time window was 10.0382 m. The percent difference is about 0.38 % for this time window.

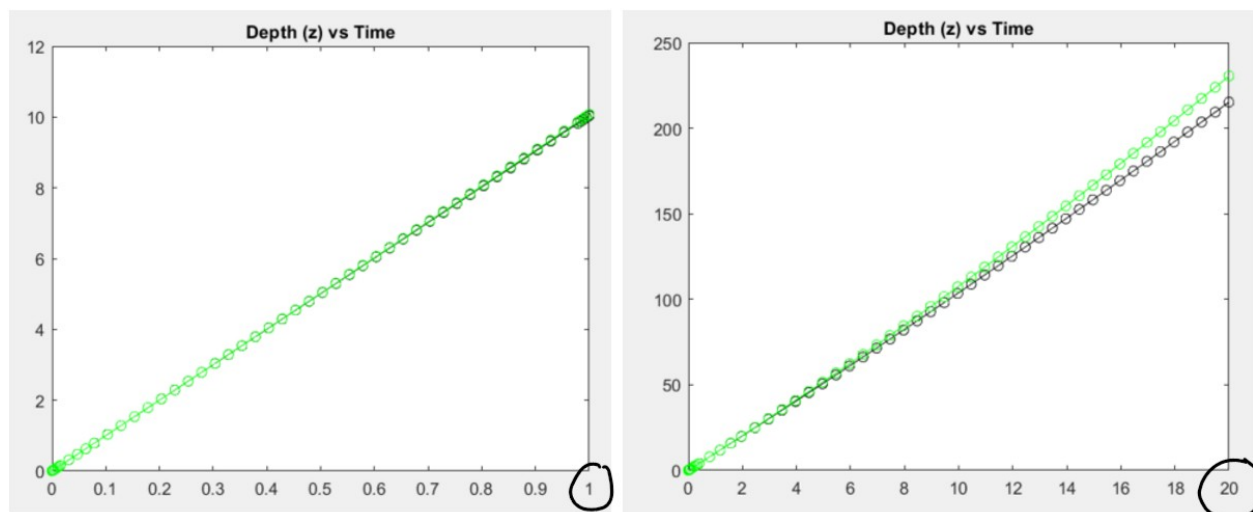
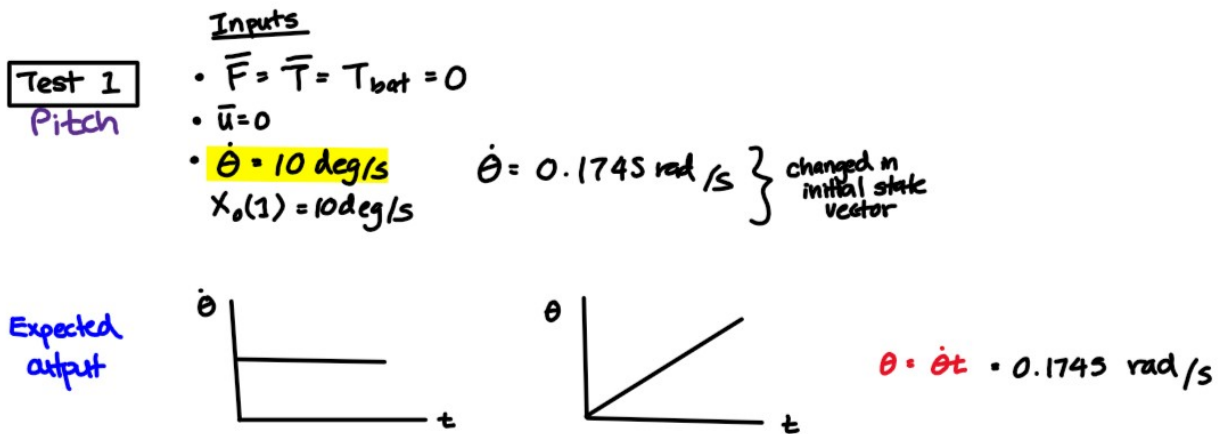


Figure 19: Same test but with longer time windows (1 sec vs. 20 sec). Depth vs. time profile.

However, as we increase the time window from 1 to 20 seconds, the simulated depth (black line) kept diverging from the expected depth (green line). It starts diverging approximately at 0.5 seconds. So, the key takeaway here is that shorter time windows should be taken when implementing ODE function.

3.2 Rotation in all directions using dynamicsV6_06.m

3.2.1 Test 1: Constant pitch rate w/ no acceleration



For this test a constant pitch rate was applied to observe how pitch behaves. When these inputs were applied, the total torque vector was not zero. It had some torque due to center of buoyancy being located some distance away from the center of gravity. In order to make it zero for this test, since we do not want any acceleration, this torque was zeroed out.

$$T_{eq} = \begin{bmatrix} J\Omega + (r_p \times \dot{p}_p) \\ 0 \\ 0 \\ -0.2132 \times 10^{-9} \end{bmatrix} \times \Omega + \cancel{(M\dot{r}_b + \dot{p}_p) \times \dot{r}_b} + \cancel{m_{bat} \cdot (r_p \times g_b)} + m_s \cdot (r_s \times g_b) + \cancel{T_{hydro}}$$

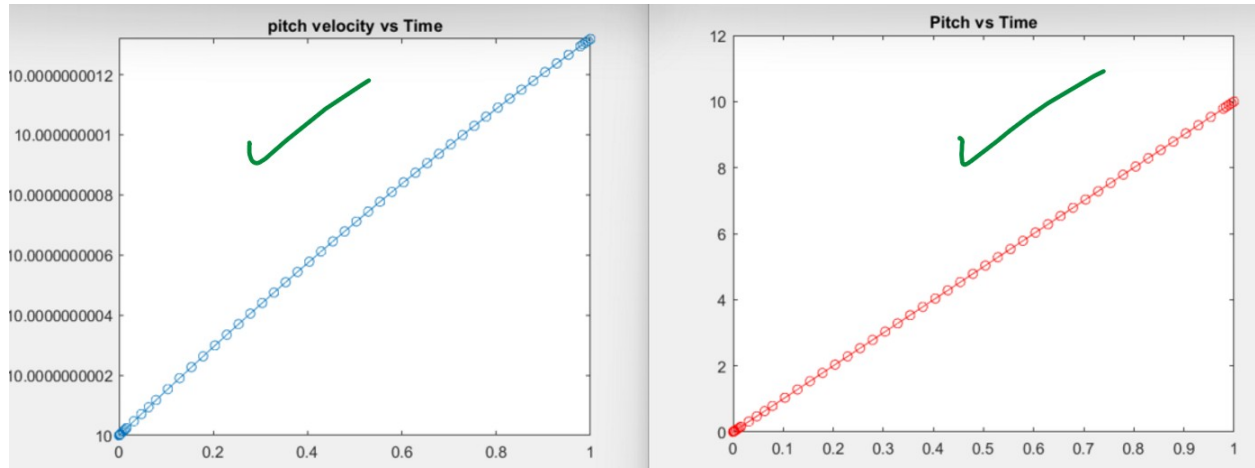
$\begin{bmatrix} 0 \\ 5.799 \\ 0 \end{bmatrix}$

new to dynamicsV6 due to CB being located away from CG
★ This was removed to proceed w/ analysis

$$T_{eq} = \begin{bmatrix} 0 \\ 5.79 \\ 0 \end{bmatrix}$$

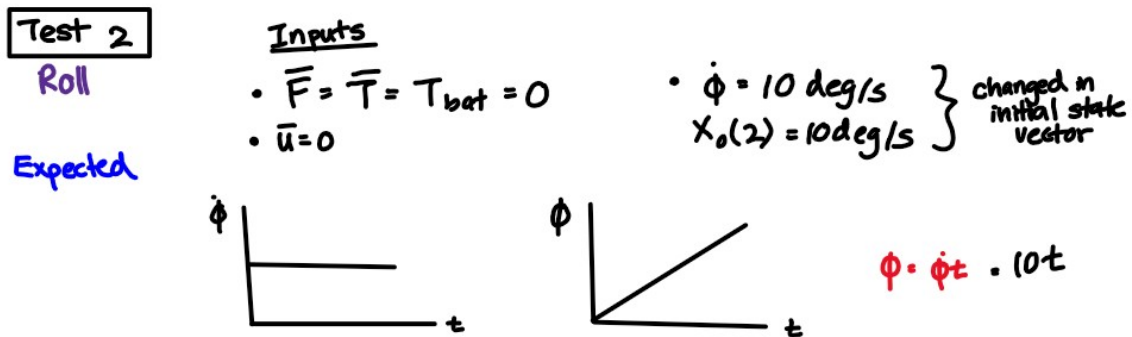
Figure 20: Torque contribution from center of buoyancy.

After this was removed, the test was run again and the following are the actual outputs for 1 second window.



The plot on the left shows the pitch rate vs. time. This shows a linear plot; however, the vertical axis shows values that are 10 deg/s. Thus, it can be concluded from this plot that pitch rate stays constant. The plot on the right shows the pitch vs. time where pitch increases linearly. This is expected as we apply a constant pitch rate. Another observation is that the red line which is the expected and black line not seen on the plot which is the actual, aligns on top of each other.

3.2.2 Test 2: Constant roll rate w/ no acceleration



For this test a constant roll rate was applied to observe roll behavior.

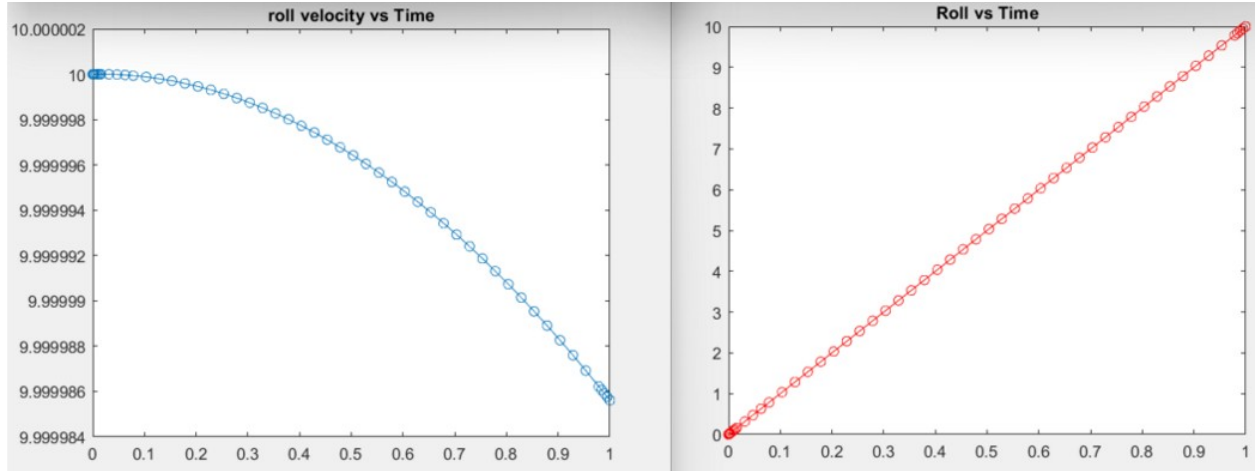


Figure 21: Test run for 1 second time window.

When test was run for 1 second time window, the roll velocity stayed pretty constant around 10 deg/s and roll angle increased linearly to 10 deg. This matches the expected results. However, it was observed that roll rate is sinusoidal with very small amplitude.

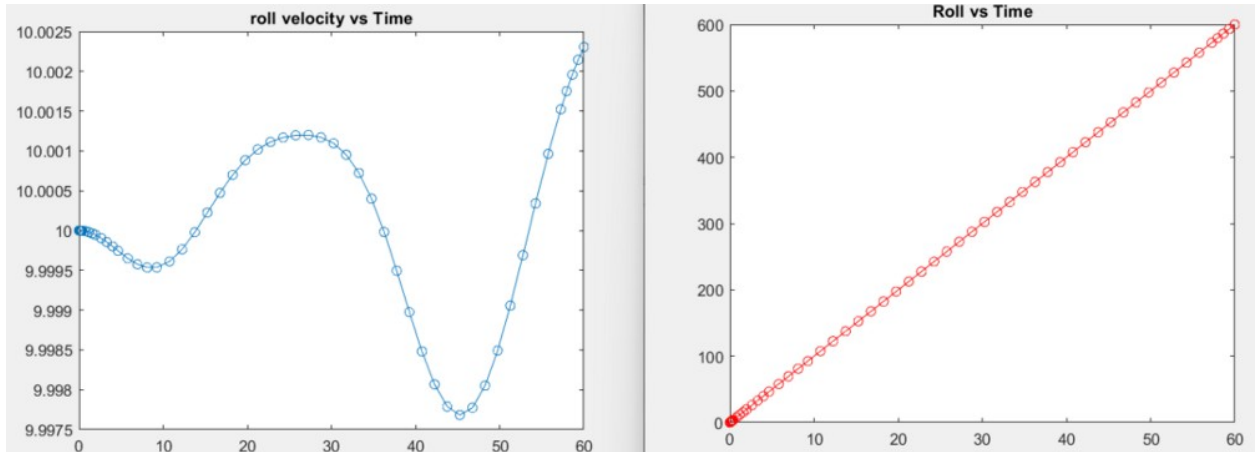
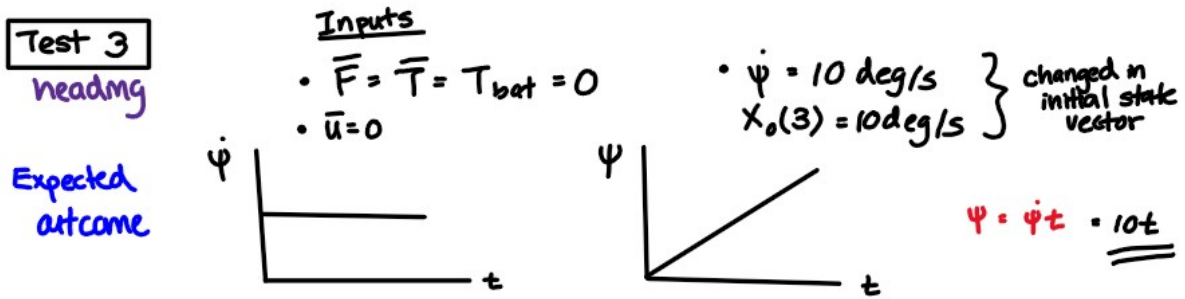


Figure 22: Test run for 10 second time window.

When the same test were run for 10 second window, the oscillations in the roll rate were more visible. However, the roll angle was observed to be behaving as expected, thus, this test passed.

3.2.3 Test 3: Constant heading rate w/ no acceleration



For this test a constant heading rate was applied to observe heading behavior.

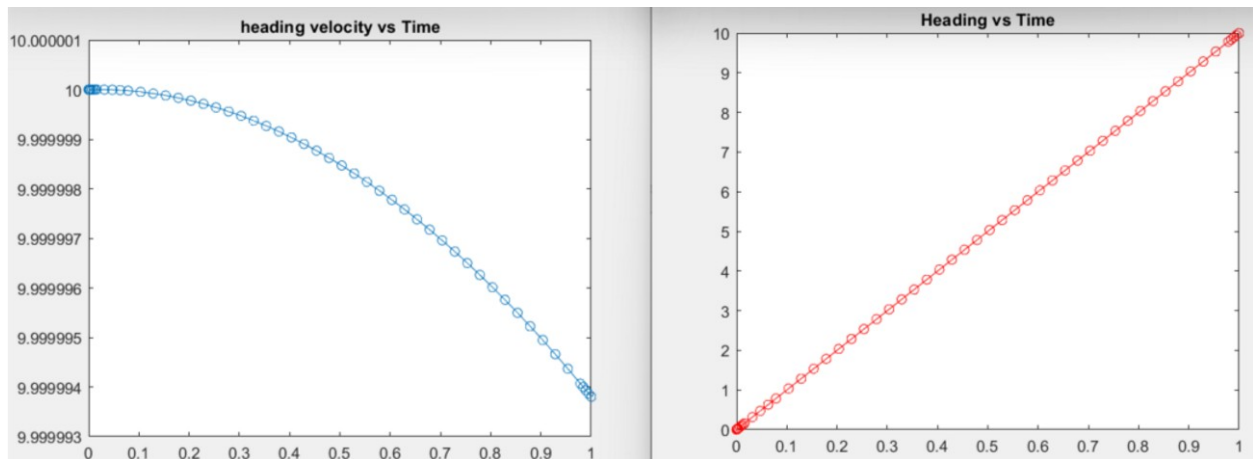


Figure 23: Test run for 1 second time window.

When test was run for 1 second time window, the heading velocity stayed pretty constant around 10 deg/s and heading angle increased linearly to 10 deg. This matches the expected results (red line and black line on the right plot aligns). However, it was observed that heading rate is sinusoidal with very small amplitude.

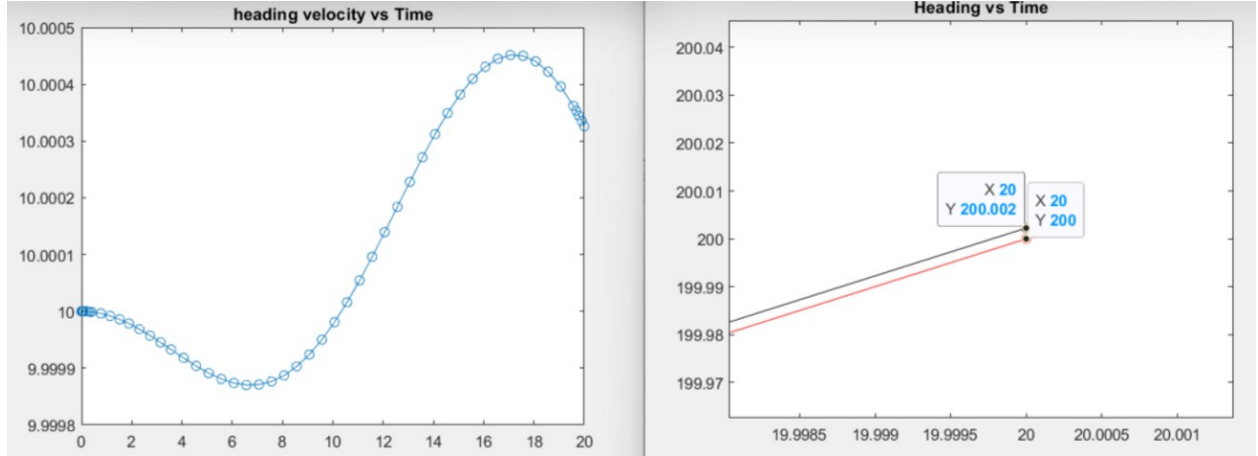


Figure 24: Test run for 10 second time window.

When the same test were run for 10 second window, the oscillations in the heading rate were more apparent with higher amplitudes, but the value is approximately 10 deg/s. However, the heading was observed to be behaving very similar to the expected, since the the percent difference is about 0.001%. Therefore, this test also passed.

4 References

1. Bennett, James S., Frederick R. Stahr, Charles C. Eriksen, Martin C. Renken, Wendy E. Snyder, and Lora J. Van Uffelen. *Assessing Seaglider Model-Based Position Accuracy on an Acoustic Tracking Range*. Journal of Atmospheric and Oceanic Technology 38.6 (2021): 1111-1123. <https://doi.org/10.1175/JTECH-D-20-0091.1> Web.